

Chikungunya transmission and susceptibility in CearC!, Brazil

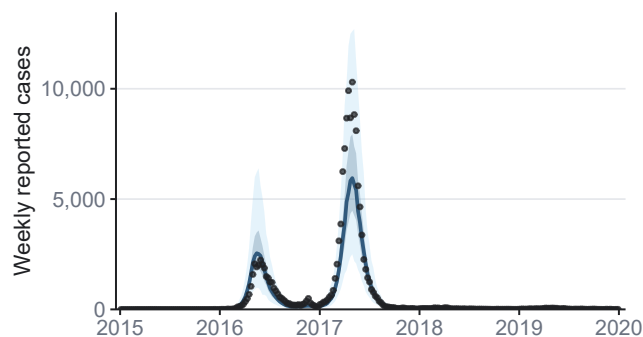
EXPLORATORY ONLY b

A

Reported cases and posterior prediction

Shading: 95% predictive interval; dark band: 50% CrI

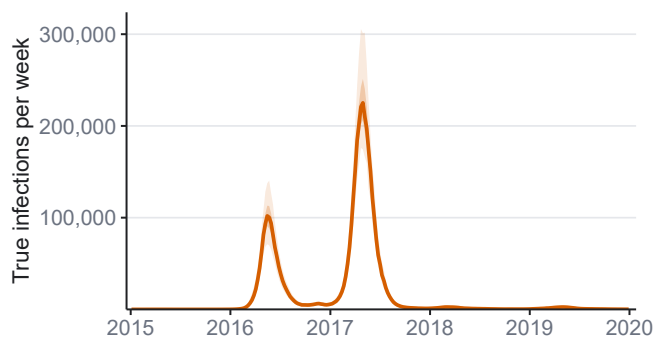
— Model expectation • Observed cases



B

Latent infection incidence

CearC! total: median, 50% and 95% credible intervals

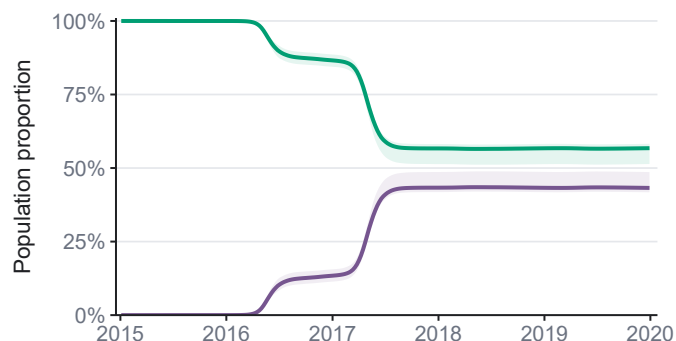


C

Population susceptibility and immunity

Population-weighted CearC! total

— Infection-derived immune — Susceptible

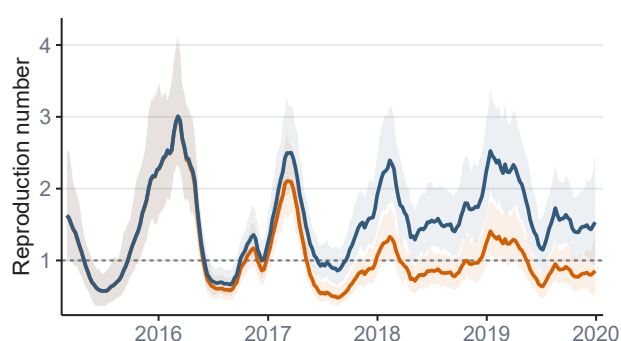


D

Transmission potential

Shared R0 trajectory and population-weighted CearC! Reff(t)

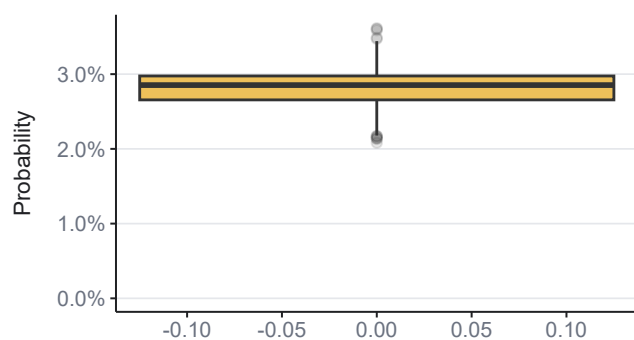
— CearC! Reff(t) — Shared R0(t)



E

Pooled infection-to-report probability

$q = 0.029$ [0.023, 0.032]

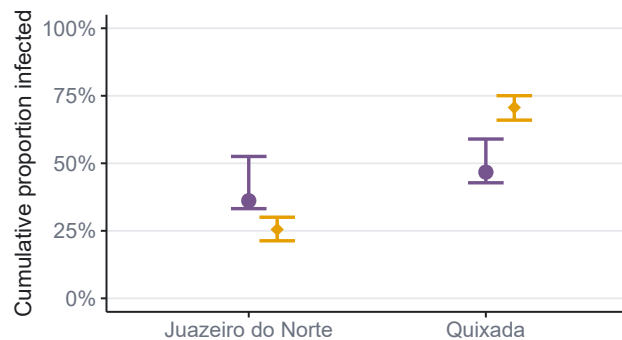


F

Direct local serology validation

Model: regional survey-window mean; bars: 95% interval

— Model site seroprevalence — Observed seroprevalence



Sub-spatial susceptible trajectories

$S(t)/N(t)$ at the start of each week; median, 50% and 95% credible intervals

